Girls Build Bots — Junior Cybersecurity Safeguards

Badge-aligned workshop guide for robotics-team-led outreach

Level: Juniors (Grades 4–5) · **Badge:** Cybersecurity — Cybersecurity Safeguards **Series:** Cybersecurity (Basics · Safeguards · Investigator) **Format:** One 90–120 minute session · **Ratio:** 1 mentor per 4–5 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on printed materials and free browser tools.

> **Troop Leaders:** This curriculum is designed to *align with* the Junior Cybersecurity badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: being safe and smart online

The series builds real digital literacy: understand how information travels (Basics), **protect it** (Safeguards), and investigate critically (Investigator). Juniors can learn concrete, practical protections they'll actually use.

Vocabulary: password, passphrase, phishing, red flag, privacy setting, digital footprint, two-factor (a second lock).

Cybersecurity Safeguards (90–120 minutes)

Badge theme: Practical protections — strong passwords, privacy awareness, digital footprints, and spotting phishing red flags.

Materials

- **Made-up** password examples for the strength lab (never real passwords — say this rule out loud)
- Mentor-made "inbox" of printed example messages: some legit, some phishing (fake prize, fake "your account is locked," a too-good-to-be-true offer)
- "Digital footprint" scenario cards (a photo posted, a comment made, a quiz taken, a location shared)
- Optional: Interland "Tower of Treasure" for reinforcement

Activity 1 — Password Strength Lab (25 min)

Sort made-up passwords weakest to strongest. Discover the rules by pattern: **longer beats clever**, mixed characters help, and a **passphrase** (three random words: "PurpleRobotTaco!") is both strong and memorable. Each Scout invents a made-up passphrase using the method —

and does NOT write down or share any real one. Mention the second lock: some accounts add a code sent to a grown-up's phone (**two-factor**) — two locks beat one.

Activity 2 — Spot the Phish (30 min)

Deal the mentor-made "inbox." For each message, teams hunt **red flags**: urgent pressure ("act NOW!"), prizes for nothing, misspelled sender addresses, links that don't match, requests for personal info or passwords. Score a point per correctly-sorted message, with the rule of thumb: **real companies don't ask for your password in a message.** When unsure — don't click; ask a trusted adult.

Activity 3 — Footprints in the Sand (20 min)

Scenario cards: for each action (post a photo, take a fun quiz, share a location), ask — who can see this? How long does it last? Would you want it visible in five years? Juniors grasp the honest version: things shared online can persist and travel. The safeguard: **think before you post, and use privacy settings with a grown-up's help.**

Activity 4 — My Safeguard Kit (10 min) + Wrap (5 min)

Each Scout writes her personal top-three safeguards (a passphrase method, a phishing red flag she'll watch for, a think-before-posting rule). Closing: pairs quiz each other with one made-up message — phish or legit?

> **Robot moment (optional):** Share for 60 seconds how the team protects its accounts and code — strong passphrases and never sharing logins — because losing control of the robot's code would be a very bad day.

Completion Checklist — Cybersecurity Safeguards

By the end of the session, to earn the badge, each Scout can: - [ ] Explain what makes a password strong and build a made-up passphrase - [ ] Spot at least three phishing red flags in example messages - [ ] Explain what a digital footprint is and why it lasts - [ ] Name her personal top-three safeguards

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Mentor Guide (for the robotics team)

- **You don't need a robot for this.** Printed inboxes and scenario cards carry the session. The "robot moment" is a 60-second add-on.
- **Never touch real passwords.** All examples invented; state the rule aloud so Scouts learn not to share real ones with anyone — including you.
- **Make phishing examples age-real.** Game-currency scams and prize DMs are their world; bank scams are not.
- **Reasoning over right answers.** Always ask "which red flag tipped you off?"
- **Calm, capable tone.** The message is "you can protect yourself," not "the internet is dangerous."

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Junior Cybersecurity badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.